

Project 7: passive DI box manufacturing proposal

PDI-12 | Revision A | 5 October 2026 | costs shown in South African rand (ZAR)

A passive transformer-isolated DI for active instruments, keyboards and unbalanced line sources. It provides an insulated TS input, parallel TS thru, nominal -20 dB input pad, ground lift and balanced male XLR microphone output. It needs no power supply. A premium transformer, restrained wiring and mechanically supported panel parts prioritise reliable audio and repairability.

Release status: complete editable design and prototype/RFQ package. This is not a tested or certified production product. KiCad PCB DRC and schematic-to-PCB connectivity checks passed; no physical unit has been built. Audio, switch orientation, mechanical fit and phantom-power behaviour must be qualified on a first article before batch production.

Deliverable files

Item	Files
Editable electronics	DI_Box.kicad_pro, .kicad_sch, .kicad_pcb; local symbol and footprint libraries
PCB fabrication	Gerbers, plated and non-plated drill files, drill maps
Mechanical	Base, lid, insulated carrier and assembly STEP; cutout DXFs; dimensioned PDF
Assembly and procurement	BOM.csv, Harness.csv, THT_insertion.csv, assembly guide
Verification	Exported netlist, DRC report, independent parity and calculations, conservative envelope checks

Requirements and intended use

Requirement	Design basis / qualification target
Power	Passive; no battery or phantom-power circuit
Load	Balanced microphone preamp, nominal 1 kOhm; qualify from 1 kOhm to 3 kOhm
Input impedance	Bypass: load- and frequency-dependent, about 140 kOhm at 1 kHz with 1 kOhm output load. Pad: about 10.08 kOhm
Gain	Nominal turns-ratio loss -21.58 dB; actual insertion loss includes winding loss and loading
Audio target	Bypass: 20 Hz-20 kHz within +/-1 dB of 1 kHz at +4 dBu, 50 Ohm source, 1 kOhm load
Distortion target	THD+N <=0.1% at 20 Hz/+4 dBu; <=0.01% at 1 kHz/+4 dBu under same fixture
Operating targets	0-50 degC, indoor dry use; no ingress rating after machining
Input limits to qualify	Bypass +18 dBu RMS; pad +28 dBu RMS; audio signals only, no speaker outputs or DC

Circuit decisions and calculations

T1 is a Jensen JT-DB-E wire-lead DI transformer. Its nominal ratio is 12:1; the datasheet gives a typical measured turns ratio 11.52:1. No substitute is authorised without revising the design and repeating audio qualification. The input and thru are parallel before the pad. A connected amplifier therefore loads the instrument in parallel with the DI.

An ideal 12:1 transformer gives $V_{sec}/V_{pri} = 1/12$ and $20\log_{10}(1/12) = -21.58$ dB. A 1 kOhm secondary load reflects as $12^2 \times 1000 = 144$ kOhm. This relation transforms an external load; it does not give a fixed input impedance or turn a quoted input impedance into a guaranteed output impedance. Real winding resistance, inductance, frequency and source impedance matter.

The manufacturer specifies approximately 141 kOhm input impedance and 150 Ohm output impedance in its stated test circuit. Its test source resistance is 6.81 kOhm. These are transformer test data, not measurements of this completed box. With a low-impedance source, this design should be evaluated using the actual source/load fixture specified above.

Pad network: R1 = 9090 Ohm series; R2 = 1000 Ohm shunt. SW1 switches both the signal selection and shunt connection. BYPASS closes 2-1 and 5-4: primary hot receives the input directly and the shunt lower end is open. PAD closes 2-3 and 5-6: primary hot receives PAD_OUT and R2 is grounded. Thus the shunt does not load the instrument in bypass.

Calculation	Result
R2 141 kOhm	992.96 Ohm
Pad voltage ratio at input jack	0.098479
20log10(voltage ratio)	-20.133 dB
Pad input impedance	10082.96 Ohm
1% resistor corners and 130-150 kOhm transformer impedance	-20.294 to -19.974 dB
Incremental pad with 50 Ohm source	-20.173 dB relative to bypass
R1 power at +28 dBu input	0.0339 W; 0.25 W part selected

At 1 kOhm source impedance, the calculated incremental pad is about -20.89 dB; at 10 kOhm it is about -25.52 dB. The pad is intended for active/line sources. Passive pickups may experience audible loading, particularly with a low-impedance thru amplifier. This is not a 1 MOhm active DI and is unsuitable for high-impedance piezo sources without a buffer.

Grounding, isolation and robustness

XLR pin 2 is hot (T1 RED), pin 3 is cold (BRN). XLR pin 1 and shell stay connected to chassis in both switch positions. T1 WHI goes to that chassis network. T1 ORG, BLK and GRY go to instrument signal ground. SW2 grounds that instrument network in GROUND and disconnects it in LIFT. The switch never interrupts the XLR shield-to-chassis connection.

The GRY lead is bonded to the transformer can. The can therefore must not touch the enclosure, metal clips or grounded fasteners. A 6 mm acetal carrier and 6 mm nylon supports isolate it. Two #4 Type B screws fasten the transformer from below the carrier; 9.525 mm screw length minus 6 mm carrier and 0.50 mm washer gives 3.025 mm penetration. Inspect the stack before installation; the vendor limit is 3.81 mm maximum.

Panel-mounted connectors and switches carry plug and toggle forces. The board mounts on four 6 mm nylon supports. All unused switch contacts are covered. Wire pairs are twisted, restrained and kept clear of the lid and sharp edges. Heat shrink and nylon clips provide solder-joint strain relief. Bonding areas are left bare under washers; conductive lid contact is checked after coating.

The balanced secondary floats; there is no intentional audio-pin-to-chassis DC path. Steady common-mode phantom power should not drive a properly balanced winding, but plug transients and wiring faults require first-article testing. No mains isolation, protective-earth function or safety certification is claimed. Never lift the protective earth of any connected powered equipment.

Standards and process intentions

The pin 1-at-chassis topology follows established professional audio grounding practice. It is a design intention, not an AES48 conformity declaration. Specify IPC-A-610 Class 2 workmanship as the contract acceptance objective, lead-free process and supplier material declarations for relevant market requirements. EMC, regulatory labelling and environmental compliance need assessment for the intended market; none is certified by these files.

Verification completed

Check	Result
Native KiCad schematic parse and netlist export	Passed using KiCad 7.0.11
Independent schematic netlist check	Passed: 48 connected pins against explicit harness specification
PCB-schematic net parity	Passed: 26 connected PCB pads
KiCad PCB DRC	0 violations; 0 unconnected pads; no exclusions
Mechanical checks	Base/lid/carrier valid solids; no overlaps among conservative part envelopes
Schematic ERC	Not run here: KiCad 7 CLI lacks ERC. Run KiCad Inspect > Electrical Rules Checker
Bench audio, real fit, phantom and durability	Not performed; first article required

Outsourced manufacturing and cost

Fabricate the board from the supplied Gerbers and both drill files: two-layer FR4, 1.6 mm +/-0.16 mm, 1 oz copper per side, lead-free HASL, green solder mask and white top silk. Design traces are 0.40 mm; board minimum clearance 0.30 mm. Finished solder holes are 1.0 mm, resistor holes 0.8 mm, vias 0.4 mm and mounting holes 3.2 mm. No stencil or SMT placement service is needed.

Machine the base from 6061-T6 aluminium. Its 150x100x50 mm body has 3 mm walls and 5 mm floor, with integral lid bosses and R3 pocket corners suitable for a 6 mm end mill. Machine the 3 mm lid and 6 mm acetal carrier separately. The STEP models show tap-drill cylinders; drawing callouts specify M3 threads and depths. Deburr 0.2-0.5 mm and remove chips before electronics assembly.

The chosen custom CNC case makes the RFQ geometry unambiguous and avoids assuming a stock casting is dimensionally interchangeable. It is expensive for a basic DI. For a commercial volume product, a suitable die-cast case would reduce machining cost, but substitution needs new mechanical drawings, clearance checks and first-article approval. The transformer is also a major cost driver; a cheaper transformer requires a complete audio requalification.

Cost basis: ZAR planning allowances for a nominal 100-unit quotation, dated 5 October 2026. The original USD allowances were converted at **1 USD = 16.5619 ZAR** (mid-market rate, 6 October 2026, 09:41 UTC). These are budget estimates, not supplier quotations, guaranteed prices, stock confirmations or landed costs. Obtain supplier RFQs before procurement. Freight, tax, duties, regulatory testing and non-recurring engineering are excluded. The rate moves daily, so re-convert before relying on these figures.

Cost category	Allowance per unit (ZAR)
Purchased/custom parts and packaging	3,625.23
Manual assembly allowance	248.43
Routine electrical/audio test allowance	132.50
Manufacturing subtotal	4,006.16
10% contingency	400.63
Budget unit total	4,406.79
100 units, excluding setup and excluded costs	440,677.38

Quote PCB fabrication, enclosure machining/coating, harness assembly and audio test as separate cost items. Request quantity breaks at 1, 10 and 100 units and identify setup charges separately. Do not order a batch until the first article passes the acceptance plan.

First-article qualification and unit acceptance

Test with calibrated instruments. Use a sine source with 50 Ohm output resistance and a measured 1.00 kOhm balanced load across XLR 2-3. Use a high-impedance differential analyser across that load; do not ground either output pin. Measure input at the input jack and output across 2-3. State source level, frequency, load, pad and lift positions for every recorded result.

Test	Method and provisional acceptance
Visual / workmanship	Correct parts and orientation; no loose hardware, burrs, wire pinch, conductive can contact or poor solder joints
Input/thru continuity	Tip-tip and sleeve-sleeve <1 Ohm, excluding probe resistance
Chassis bond	XLR 1 and shell to enclosure <0.1 Ohm using a four-wire or corrected low-resistance method; both switch states
Ground lift	GROUND: sleeve-chassis <1 Ohm. LIFT: >20 MOhm with DMM; transformer can-sleeve continuous, can-chassis isolated
Winding check	YEL-ORG about 4.6 kOhm; RED-BRN about 63 Ohm. Record actual; vendor values are typical, not production tolerance limits
Isolation	Input network to output pins >20 MOhm in LIFT using DMM. Do not perform undocumented high-voltage testing
1 kHz gain / polarity	Bypass loaded gain provisional -20.5 to -23.5 dB. Positive input produces positive XLR 2 relative to 3
Pad increment	At 1 kHz, 50 Ohm source and 1 kOhm load: -20.2 dB +/-0.5 dB relative to bypass

Test	Method and provisional acceptance
Frequency / distortion	First article: full 20 Hz-20 kHz sweep and THD+N at 20 Hz/1 kHz. Limits as requirement table; fixture noise below limit
Maximum-level qualification	Verify provisional +18 dBu bypass and +28 dBu pad at 20 Hz. THD+N <=1%, no overheating; test and revise if needed
Phantom compatibility	First article: controlled 48 V common-mode phantom with matched 6.81 kOhm feeds; check transient behaviour and absence of source-side DC
Mechanical durability	First article: repeated jack insertion/toggle operation, cable restraint and lid fit; then re-run bonds, lift and audio
Routine batch tests	Visual, continuity, bonds, lift, isolation, 1 kHz gain/polarity, pad; lot-sample sweep/THD and all units after process changes

After first-article qualification, replace provisional limits with measured, justified production limits and approved inspection fixtures. Record serial number, lot, operator, instrument IDs, measured values, pass/fail and rework. Re-test after every repair. These files cannot provide laboratory evidence without a physical unit.

BOM summary and reference register

Allowances are per line (quantity included) in ZAR, converted from the original USD allowances at 1 USD = 16.5619 ZAR. Each line is rounded to the cent, so the sum of the rounded lines can differ from the total by a few cents.

Reference	Part / specification	Qty	ZAR allowance
T1	JT-DB-E	1	1,656.19
J1,J2	NMJ2HF-S	2	99.37
J3	NC3MD-L-1	1	115.93
SW1	7201SYZBE	1	132.50
SW2	7101SYZBE	1	115.93
R1	MFR-25FBF52-9K09	1	1.66
R2	MFR-25FBF52-1K	1	1.66
PCB1 / H1-H6	PDI12-PCB-A	1	49.69
ENC1	PDI12-BASE-A	1	1,076.52
LID1	PDI12-LID-A	1	149.06
CAR1	PDI12-CARRIER-A	1	82.81
ST1-ST4	M3-NYLON-MF-6-MALE3	4	16.56
ST5-ST8	M3-NYLON-MF-6-MALE3	4	16.56
F1-F4	ISO7045-M3x4-A2	4	3.31
F5-F8	ISO7045-M3x8-A2	4	3.31
F9,F10	ANSI-TYPE-B-N4x3/8-PAN	2	3.31
W1,W2	N4-WASHER-T0.5	2	1.66
F11-F14	ISO7045-M3x8-A2	4	3.31
F15-F17	ISO7045-M3x8-A2	3	2.48
N1-N3	DIN985-M3-A2	3	3.97
W3-W6	DIN6798A-M3-A2	4	3.31
LUG1	M3-RING-22AWG	1	3.31
WIRE1	WIRE-24AWG-STRANDED-300V	1	16.56
WIRE2	WIRE-22AWG-STRANDED-300V	1	3.31
INS1	PDI12-INSULATION-KIT	1	16.56
FEET1-FEET4	RUBBER-FOOT-D10-H3	4	13.25
PACK1	PDI12-PACK-A	1	33.12
Total			3,625.23

Opening the project and controlling the release

Extract the ZIP into one folder and open DI_Box.kicad_pro in KiCad 7 or newer. Open the schematic and PCB from the project manager. Project-local symbol and footprint libraries are included and use \${KIPRJMOD}; no downloaded commercial CAD libraries are required. Newer KiCad versions may offer to migrate the files: work on a copy if preserving this release.

Before manufacturing: run schematic ERC and PCB DRC in the installed KiCad version, review the schematic-to-PCB changes dialog, inspect Gerbers and both drill files in a Gerber viewer, obtain supplier drawings for purchased parts, check delivered switch terminal numbering/orientation and perform first-article qualification. Export a revised manufacturing package if any circuit, footprint or geometry changes. Keep BOM, harness and drawings at the same revision.

This design uses no SMT parts; THT_insertion.csv is a manual insertion reference. It is not a conventional SMT pick-and-place file. H1-H6 are labelled wire-solder holes rather than plug-in headers. The PDF schematic is a net-labelled manufacturing schematic: named nets define continuity between circuit blocks and terminal pads.

Manufacturer source register

Jensen JT-DB-E datasheet, winding colours / can connection / mounting and transformer test data

<https://www.jensen-transformers.com/wp-content/uploads/2014/08/jt-db-e.pdf>

Neutrik NC3MD-L-1 product and drawing, XLR cutout and mount pattern

<https://www.neutrik.com/en/product/nc3md-l-1>

Neutrik NMJ2HF-S drawing, jack cutout and body envelope

<https://www.neutrik.com/media/8572/download/st-nmj2hf-s.pdf?v=2>

Littelfuse C&K 7000 series, switch configuration and terminal scheme

<https://www.littelfuse.com/products/switches/toggle-switches/7000>

Yageo leaded-resistor family; obtain current MFR series drawing with order

https://www.yageo.com/en/Product/index/product_category/resistor

KiCad 7 PCB documentation and Python API

<https://docs.kicad.org/7.0/en/pcbnew/pcbnew.html>

Sources checked 5 October 2026. Purchased-part drawings remain the authority for those parts. New enclosure geometry and PCB are original to this proposal; manufacturer application drawings are not redistributed. Currency conversion rate source: mid-market USD/ZAR, 6 October 2026.